

Sustained Selected Complex Receive Streaming from the BCM43455 on Raspberry Pi 5

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Author: Adam Davies. Source and evidence: project repository. Report and diagrams: CC BY 4.0; original code: GPL-2.0-or-later. Retained upstream terms apply separately.

Abstract

We demonstrate a checked one-hour recording of selected radio receive samples from the Raspberry Pi 5's onboard BCM43455c0 Wi-Fi chip. Patched manufacturing firmware fills small buffers while a Linux driver reads them and one hardware collector remains active. In accepted recordings, every requested output passes sequence and checksum checks, with controlled stop and checks for stalled readers. The output intentionally retains selected or partly filtered samples at reduced rates, rather than exporting the complete input. In a separately audited five-minute STEP0 recording, two short known-source bursts produce detectable signals: all 80 predeclared test chunks pass. Included source and build instructions reproduce the exact tested firmware and driver bytes. These results establish one-hour bounded sample delivery, observed raw-position rollover and separate early and late source correlation. They do not establish complete-input physical continuity, calibrated RF bandwidth or comprehensive anti-alias filtering. The operational controller remains private, so the hardware experiment is not turnkey.

1 Purpose and scope

Normal Wi-Fi software exposes decoded packets. Time-domain receive samples let software examine the received waveform without requiring a decoded packet for each exported block. This project demonstrates an experimental radio-sample streaming capability using the Raspberry Pi 5's existing BCM43455c0 Wi-Fi chip, with an audited one-hour recording.

This project was inspired by ESP-SDR's sample access on Espressif chips, including its continuous streaming implementations. The Broadcom chip already contained a finite diagnostic sample-capture mechanism. Our contribution is adapting it into sustained, checked export to Linux on this particular chip and firmware, while the collector remains active and small buffers are reused. We establish no first-ever radio-sample-access claim. The pinned comparison and section 9 identify inspected precedents.

The essential idea is to read small buffers before they are reused. Firmware takes completed values from the chip's circular sample memory and publishes blocks in a second reusable buffer. A Linux reader retrieves those blocks while collection continues. Sequence, checksum and lifecycle checks distinguish accepted delivery from an incomplete or failed recording.

The stream is intentionally reduced-rate. Ordinary capture produces nominally 20 million packed complex words per second; selecting one input position in sixteen gives nominally 1.25 million output words per second. Other tested configurations combine several inputs before selection and deliver nominally about 833,333 output words per second. These differences express intentional selection or partial filtering, not transport loss.

A separate five-minute source demonstration records two short, bounded ESP32-C3 bursts, one near the start and one near the end. The later passive one-hour trial extends delivery validation and contains no deliberate source operation. All 80 predeclared STEP0 RF test chunks pass their fixed criteria. This supports early and late source correlation during one receiver session; the transmitter is not active for the full five minutes.

Access to radio samples and a practical general-purpose receiver require different evidence. Capture mode 17 is identified with the vendor-labelled `rx_filt_1core` receive path. Its exact ordering relative to all internal receive filters and corrections is uncharacterised; bare-ADC access is not established. The reduced-rate stream still needs comprehensive anti-alias filtering, calibrated tuning/bandwidth and continuous protocol decoding. Making it a practical general-purpose receiver is the next stage. Checked host sequences also do not independently locate every physical input sample time.

The evidence includes the original five-minute operating points, supporting shorter tests, later separately identified sparse32/twin6 trials and the separately audited ten-minute and one-hour passive follow-ups. The public run registry retains their configuration identities and aggregate results. This report summarises prior complete saved-archive audits and inspected implementation; publication preparation performs offline checks, not independent hardware replication. The individual experiments retain their own dates and identities. Included source rebuilds the exact tested firmware and driver bytes, but the private operational controller prevents turnkey repetition of the hardware experiment.

2 Hardware and firmware context

The tested hardware is Raspberry Pi 5 with BCM43455c0/CYW43455 onboard wireless silicon. Chip naming is retained for discovery; similarity of names does not imply compatibility between revisions or binaries. The experimental firmware is a patched manufacturing image identifying itself as 7.120.5.1 C0 WLTEST. It differs from the stock 7.45.265 firmware used for ordinary networking.

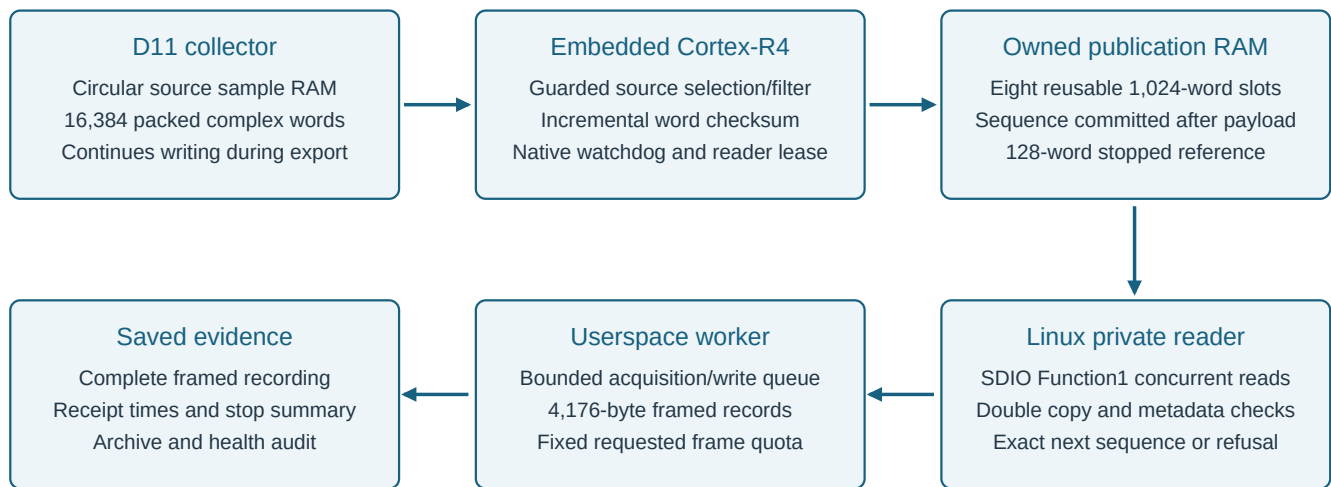
The accepted sustained baselines use numerical capture mode 17, core0, vendor downsamp=0 and a 20 MHz channel configuration on channel 40, nominally 5.200 GHz. The vendor label associates this mode with rx_filt_1core. Exact ordering relative to all receive corrections and filters is unresolved. Accordingly, the measured storage is described as packed complex receive samples, with no claim of bare-ADC access.

One 32-bit little-endian word stores two signed 16-bit lanes. The host reference convention assigns the low lane to the real component and the high lane to the imaginary component. Physical I/Q assignment and frequency sign are uncalibrated; an earlier finite Wi-Fi decoder used a conjugate convention. Sixteen-bit storage does not establish converter resolution or effective dynamic range.

Exact baseline firmware/driver identities and image-specific collector addresses are listed in the implementation appendix, subsection A1. The public run registry records the separately tested configurations.

3 Concurrent collection and publication

The hardware collector writes cyclic sample memory. Firmware observes its writer pointer, checks completion and moving lag, and reads only the inputs required for each output. The initial baselines publish blocks of 1,024 words into an eight-slot owned-RAM ring. Later accepted sparse32/twin6 trials use thirty-two slots and a different single-payload reader contract, described separately below. The host retrieves committed slots while the synchronous capture request is still outstanding. A bounded userspace queue allows acquisition to overlap buffered filesystem writes.



Schematic: native capture request stays active until normal completion

Figure 1. Source-derived architecture. Collection, firmware consumption and host reading overlap; the original command returns its genuine completion after stop. Source sample memory and host publication RAM have distinct lifetimes and capacities.

Each output slot contains 32 bytes of metadata and a 4,096-byte payload. Its sequence is invalidated before reuse. Firmware writes payload and metadata, commits the publisher checksum, executes a memory barrier, commits the slot sequence, executes another barrier and advances the published sequence. It never asks the hardware sample producer to wait for a slow host consumer.

The initial checked double-copy Linux reader verifies the expected descriptor and exact next publication. It reads slot metadata, copies the payload twice, rereads metadata and accepts only unchanged metadata, identical copies, correct publisher checksum, expected sequence/index and zero slot flags. Data frames reach userspace through bounded partial reads. A slow reader that falls beyond the output ring's recoverable window receives an error rather than silent best-effort data.

The publisher checksum is a word-wise FNV recurrence over all 1,024 packed words. It is an integrity check, not a cryptographic freshness proof. Double reads expose unstable copies but cannot prove that the collector generated fresh RF information at every nominal sample time.

4 Processing and rate budget

The ordinary selected path reads one completed source word and advances sixteen positions per output. The six-input path reads six consecutive values and applies nested signed half-add operations independently to both lanes. Its ideal linear weights are [1,1,2,2,1,1]/8. Actual integer semantics include intermediate arithmetic rounding and are specified in the appendix.

For the accepted six-input channel-40 configuration, the full PHY1a7:1a6 pair changes from 0x19ce38e3 to 0x1ab55554 through native helpers. Earlier finite production measurements associate that setting with nominal 13.333333 million input words/s, approximately one third below ordinary production. The pair is admitted before mutation, checked at arm, retained through stop/reference and restored completely afterward. This association does not establish a universal meaning for those bits or an undocumented clock-tree model.

Path	Nominal input words/s	Inputs read per output	Nominal output words/s	Sample payload MB/s
Ordinary selected	20,000,000	1	1,250,000	5.000
Modified six-input	13,333,333.333	6	833,333.333	3.333

MB denotes 1,000,000 bytes. Exporting the complete ordinary 32-bit input would require 80 MB/s; the modified input would require approximately 53.333 MB/s. The historically observed four-bit 50 MHz DDR50 link has a 50 MB/s gross ceiling. That bus mode was not remeasured during Sixstream C, and gross capacity excludes protocol costs. SD Association bus-speed overview

An exported userspace frame is 4,176 bytes: 4,096 payload bytes and 80 bytes of metadata/timing. The original V26/be069 checked SDIO path reads at least 8,384 data bytes per frame: 128 descriptor bytes, two 32-byte slot-metadata reads and two 4,096-byte payload reads. At the nominal six-input rate this is approximately 6.823 MB/s, before empty polls, lease updates, window accesses and bus overhead. Payload cadence is therefore not a maximum transport-capacity measurement.

The important acquisition limit is embedded sample-port and complete-loop service cost. The accepted six-input append distributes checksum accumulation across new words instead of rescanning a completed block at publication. It retains the 128-output batch cap and original timing/lag/ownership checks. Its maximum recorded batch cost is 20,338 counterticks; the exact maximum complete-loop duration is not exported. Copy-only budget calculations exclude other firmware work and are not total CPU-utilisation measurements.

5 Lifecycle and host control

The reader initializes an exact owned-RAM heartbeat. Firmware admits its initial token, observes expected modular token increments and services the existing nonzero native watchdog deadline. The heartbeat lease expires if renewal ceases; this is distinct from the native watchdog, which protects another part of firmware execution. Timer and pointer arithmetic use explicit modular rules rather than treating a counter wrap as ordinary increasing arithmetic.

Closing the selected stream can request STOP through one fixed owned-RAM word after identity and state checks. Firmware finishes an aligned publication block and returns through normal vendor cleanup. A separate stop summary verifies the idle descriptor, retained latest tail and stopped aperture. The producer can publish one intentional suffix block after the host's requested quota. That block is accounted for separately and need not have reached the host.

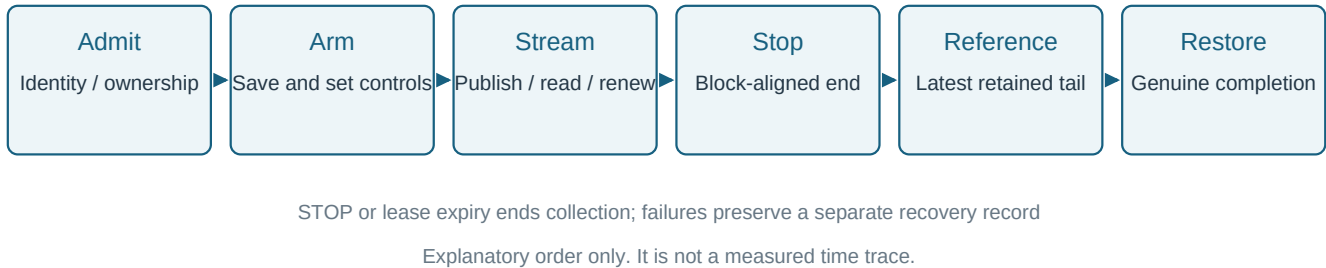


Figure 2. Explanatory lifecycle schematic, not a measured event trace. Native cleanup and completion occur after collection ends. The stopped reference uses the latest firmware-retained tail, which may belong to the intentional unread suffix.

A long capture through the earlier nl80211 control path could hold networking locks and stall unrelated progress. The accepted private module uses a privileged fixed capture endpoint outside that dispatch. It enrolls protocol-mutex users before queueing, admits one capture owner only when no users are pending, and refuses competing radio controls with EBUSY. The original command remains synchronous and its real completion reply is preserved. The data reader uses SDIO RAM access concurrently with that lifecycle.

In the original five-minute baselines, the host worker separates acquisition and recording with an 8,192-frame bounded queue, containing about 34.21 MB of frame bytes plus object overhead. The queue absorbs finite filesystem stalls; exhaustion still stops the experiment. Python buffering and kernel/storage caching mean a short write call is not evidence of durable storage completion. Host CPU, IRQ load and fsync latency were not systematically recorded.

6 Evaluation method and results

The prior audits bind tested image/module identities, frozen source packages, raw control exchanges, full frame recordings, sequence/checksum accounting, stop/reference, native restoration and retained health. The package exports allowlisted aggregates and relative receipt bins. Complete original RF and operational recordings remain private, as described in data availability.

Publication preparation additionally reruns the new offline data-frame verifier on the complete Private long B, Sixstream C and Sparse A recordings: 614,848 frames containing 629,604,352 packed words. Every frame's sequence, metadata and publisher checksum passes, and file identities are recorded in preparation checks. This repeats supplied-file integrity checking. It does not repeat the control-wire, restoration or hardware-health experiments; Lease C support remains its prior complete audit.

For F frames of N=1,024 words, delivered count is F*N. Cadence from first-to-last equivalent receipt markers uses (F-1)*N divided by their span. This is host delivery timing, not an independent physical capture duration. Expected requested counts come from fixed host quotas. Their agreement with nominal durations inferred from the same counts is not an independent continuity test.

Run	Frames	Complex words	Host receipt span s	Approximate payload cadence MB/s
Private long B selected	368,640	377,487,360	301.990723487	4.999973
Sixstream C partial prefilter	244,160	250,019,840	300.023884130	3.333319

Run	Frames	Complex words	Host receipt span s	Approximate payload cadence MB/s
Lease C selected	73,728	75,497,472	60.397535486	about 5.000
Sparse A short follow-up	2,048	2,097,152	2.515536915	about 3.333

The two five-minute configurations establish incremental delivery far beyond available on-chip storage. The collector remains active, the original capture command has not returned, eight publication slots are repeatedly reused, and checked frames arrive throughout the recording. Taken together, these observations rule out an explanation based solely on a large capture drained after completion.

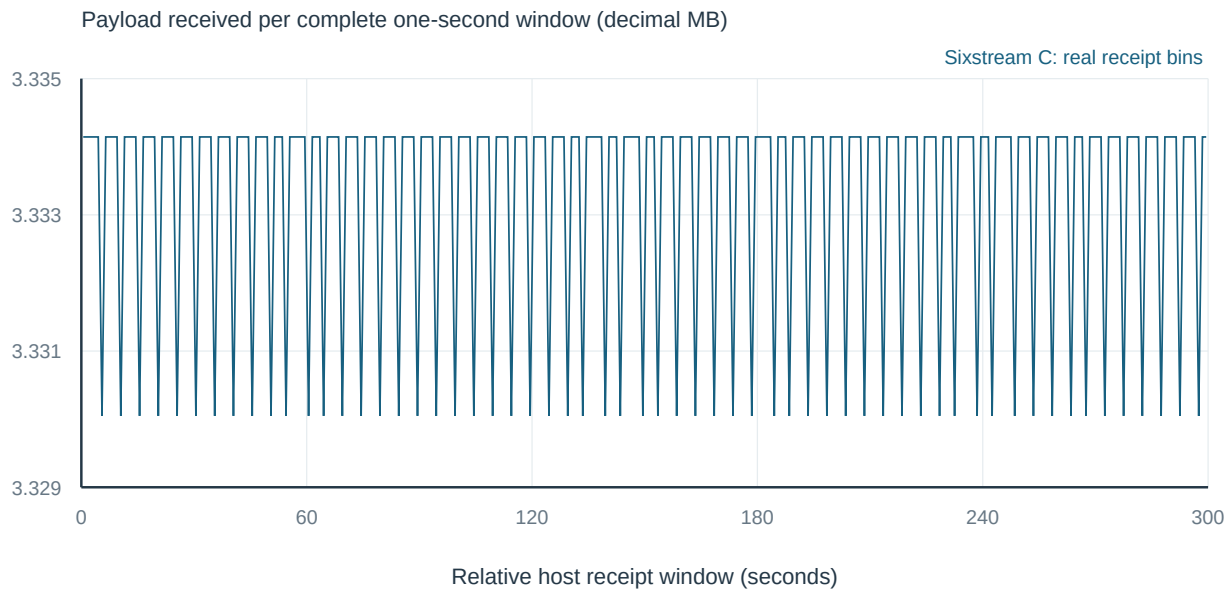


Figure 3. Real Sixstream C userspace receipt data aggregated into 300 complete half-open one-second windows. Each bin contains 813 or 814 frames, corresponding to 3.330048 or 3.334144 MB of payload. The final partial window is excluded. This describes delivery scheduling, not calibrated RF timing.

Every requested frame sequence and publisher checksum passes the prior complete audits. Sixstream C has 244,160 delivered frames and 244,161 firmware blocks, including one unread suffix. Its maximum observed lag is 2,666 raw words, with 1,907 watchdog services and lease renewals. Its recording queue peaks at 93 of 8,192 frames. The exact full-loop maximum remains unexported; guard enforcement is supported by the reviewed implementation.

Private long B records all requested frames despite a 5.678687167-second blocking write. Its queue reaches 6,931 of 8,192 frames. This is evidence that the larger queue handled that particular stall, and a warning that finite buffering has limited reserve. The quieter Sixstream C queue behaviour does not establish the same headroom for every storage workload.

7 Limitations and validity

Validation applies to the exact tested firmware, driver, capture configuration and bounded workload. Watchdog service, reader renewal, publication ownership, lag bounds, exclusive control and a bounded host recording queue are part of the operating contract. Different workloads, storage conditions or longer operation need their own validation.

Healthy retention is assessed through recorded unchanged originals, loaded/selected driver identity, Ethernet and sensor responses, boot/service continuity and inactive recovery guards. It is a finite observation under the documented workload, not an unlimited reliability guarantee.

The firmware extends a modulo source pointer with sampling/timing constraints. Those checks can detect observed excessive lag, unstable clock/context or exceeded work bounds. An independent absolute sample-generation

counter is absent. An unrecognised full lap or unmodelled collector pause cannot be excluded solely by a correct host sequence.

Earlier full-rate finite samples yield complete Wi-Fi frames with independent decoder and frame-check validation. A new offline replay independently recovers twelve complete 1 Mbps DBPSK frames from preserved channel-1 records; all twelve MAC checksums and frame residues pass, and all twelve deliberate one-bit corruption controls are rejected. This is an offline decoding result from those finite records, not a complete acquisition-session validation or a decoder for the reduced-rate continuous stream. It supports useful radio waveforms in the receive-tap family. It does not characterise every long-stream word or prove calibrated narrowband transfer. Controlled passband/blocker/alias tests and clock/gain/dynamic-range measurements remain separate research needs.

8 Partial filters and the sparse follow up

The six-input kernel reduces some high-frequency energy before stride16 but is a short partial prefilter. Its ideal model gives only weak rejection near the first folding band. A later strong host FIR can filter before a second decimation but cannot remove energy already folded into the wanted band by the first selection.

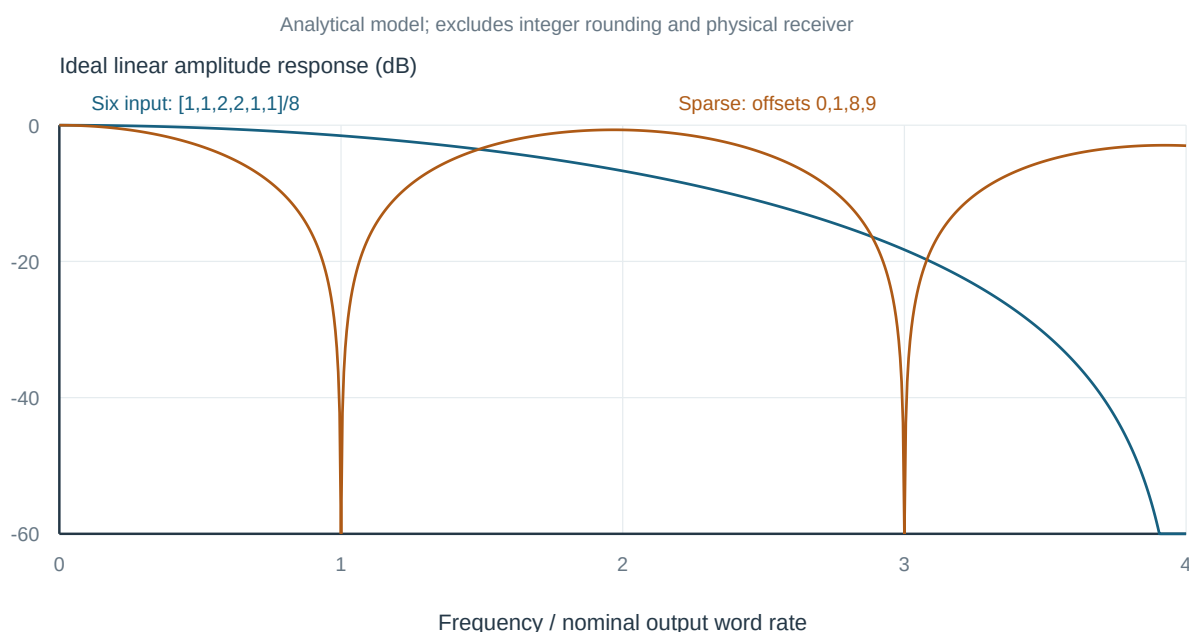


Figure 4. Analytical ideal linear responses against frequency divided by the nominal output rate. Integer rounding and the physical receiver are excluded. Sparse offsets 0,1,8,9 create odd-alias notches while leaving even-alias exposure. Mathematical zeros are not measured electrical rejection.

The independently audited sparse A follow-up uses four inputs at offsets 0,1,8,9 with signed nested averaging and the modified channel-1 production profile. Its 2,048 consecutive frames span 2.515536915 seconds. Mean active copy cost including incremental hash is 136.756814 counterticks/output, maximum batch copy 17,508 and maximum lag 2,600. Normal stop, full native pair restoration, 16 renewals and retained health pass. Separate stopped-cost epochs compare 3,072 outputs against 64,000 same-epoch raw words: 1,536 sparse-four and 1,536 six-input references. This is the combined count for that experiment.

This establishes short live feasibility, not five-minute reliability of the new filter/channel combination. The ideal sparse model improves one nearest-alias family but is worse for aggregate flat-white folding than six and leaves even aliases exposed.

A separate finite channel-1 comparison provides additional processing evidence. Its separate saved audit checks four OFF/STEP10/STEP10/OFF epochs, 64,000 raw words and all 3,072 signed-tree filter references with native control/selector restoration and retained health. The source produces multiple coherent products. Analysis jointly fits those products and DC, and retains the raw background and exact integer-floor error. Matched 256-output windows compare each kernel with the dense raw window from its own epoch.

Source-linked ON epoch	Six attenuation dB	Sparse attenuation dB	Additional sparse attenuation dB
2	1.710	18.449	16.739
3	1.878	17.695	15.817

Both target frequencies are near -0.067092 cycles/input. Their nominal folded frequencies are about -61.23 kHz, outside the wanted +/-40 kHz band. These measurements establish bounded nearby-line software-filter attenuation at the recorded tap. They do not establish wanted-band alias rejection, a general receiver low-pass response, calibrated antenna gain or an absolute clock. The 128-output windows are prefixes of the 256-output windows, not independent repetitions. Finite RF result registry

8a Later five-minute and selected-line follow-ups

These results use a separate ABI35/ABI36 thirty-two-slot owned-RAM ring and one payload copy with stable metadata and publisher checksum checks. The original eight-slot double-copy description is retained for its original operating points. A larger ring provides four times the publication retention; the changed reader reduces ordinary RAM transfer bytes by 48.85%. Because both changes were combined, the five-minute results do not isolate their individual causal benefit. Hardware sample-ring capacity and unobserved-lap limitations remain unchanged.

Later trial	Complex words	Host receipt span s	Interpretation
Sparse32 A	250,019,840	300.024784325	Sparse-four, 32 slots, single payload copy
Twin6 five-minute A	250,019,840	300.025637336	Broader partial filter, same output rate

Both deliver 244,160 consecutive frames with publisher checksums, normal STOP, one unread suffix, full native pair restoration, 128 newest-owned-tail references, 1,907 watchdog services/lease renewals and healthy retained operation. Their source identities and prior complete audits are selected explicitly in the run registry. These preparation steps reuse those completed audits; they do not repeat their hardware or full recording checks.

The twin6 partial kernel takes offsets 0,1,2,8,9,10. Its ideal weights are [1,2,1,1,2,1]/8. Separate finite matched raw epochs validate all 3,072 signed references from 64,000 raw words. Two independently started source epochs measure an additional 3.332/4.683 dB selected-line reduction relative to sparse-four, compared with an ideal 4.538 dB. The selected folded line is near +0.23338 cycles/output, outside the wanted +/-0.048 region. These are modest but physically measured selected-line differences, not a general low-pass response or a measured white-noise improvement.

8b Five-minute known-signal demonstration

A new completed experiment retains the same twin6 b4b firmware, d437 module, ABI36 and 32-slot configuration. It delivers 244,160 checked frames containing 250,019,840 complex words over 300.025188989 host seconds, with normal STOP, one unread suffix, 128 newest-owned-tail reference words, full native pair restoration and 1,907 lease/watchdog services. The saved wire/frame/archive/health review passes. This AI-assisted saved-data review reused a parser derived from a separate audit implementation; it is not independent parser authorship or independent laboratory replication.

An ESP32-C3 emits two bounded STEP0 bursts around 10 and 277 seconds while that single collector remains active. The transmitter is normally off. This establishes early and late source correlation during the five-minute receiver session; it is not a continuously transmitting five-minute source or a claim that every output word contains that signal. No receiver gain change, firmware flash or capture restart separates these bursts.

The prospective primary frequency gate is [-0.13,-0.02] cycles/output. Four ON windows contain 20 nonoverlapping 16,384-word chunks each. All 80 chunks pass the unchanged criteria, including at least tenfold amplitude over matched adjacent OFF95, three counts and three descriptive floors, plus frequency stability. Each window exceeds the minimum 12 passing chunks. The smallest individual ON/OFF95 ratio is 26.733; the four median ratios are

approximately 81.77, 76.23, 78.71 and 71.75. The selected frequency is near -0.079 cycles/output, outside the earlier wanted ± 0.048 region. This demonstrates a received source-linked line at this tap, not a calibrated wanted-band passband or additional filter attenuation. No dense raw reference is present in this recording.

Forty optional neighbor-median diagnostics are unavailable because all eligible neighbor projections were excluded near fitted nuisance lines. A strict-JSON supplement represents these as null, never zero, preserving the original NaN-bearing analysis and every RF estimate, criterion and pass flag. Operative quantities remain finite. The fallback conditional least-squares scales are descriptive: IID, stationarity and confidence coverage have not been validated. The result does not establish an empirical neighbor-floor estimate for all 80 chunks.

The sanitized predeclared plan, all 80 ON chunk fit values and matched OFF summaries, and frozen mathematical method and decision reproducer make the acceptance arithmetic inspectable. The standard-library reproducer recalculates chunk decisions, frequency stability and the minimum 26.73026989053336 ON/OFF95 ratio. OFF95 percentiles are supplied bound summaries; individual OFF fit rows were not retained in the selected response. This public check does not refit withheld waveforms, recompute those OFF percentiles or repeat source timing, capture integrity or hardware-health validation.

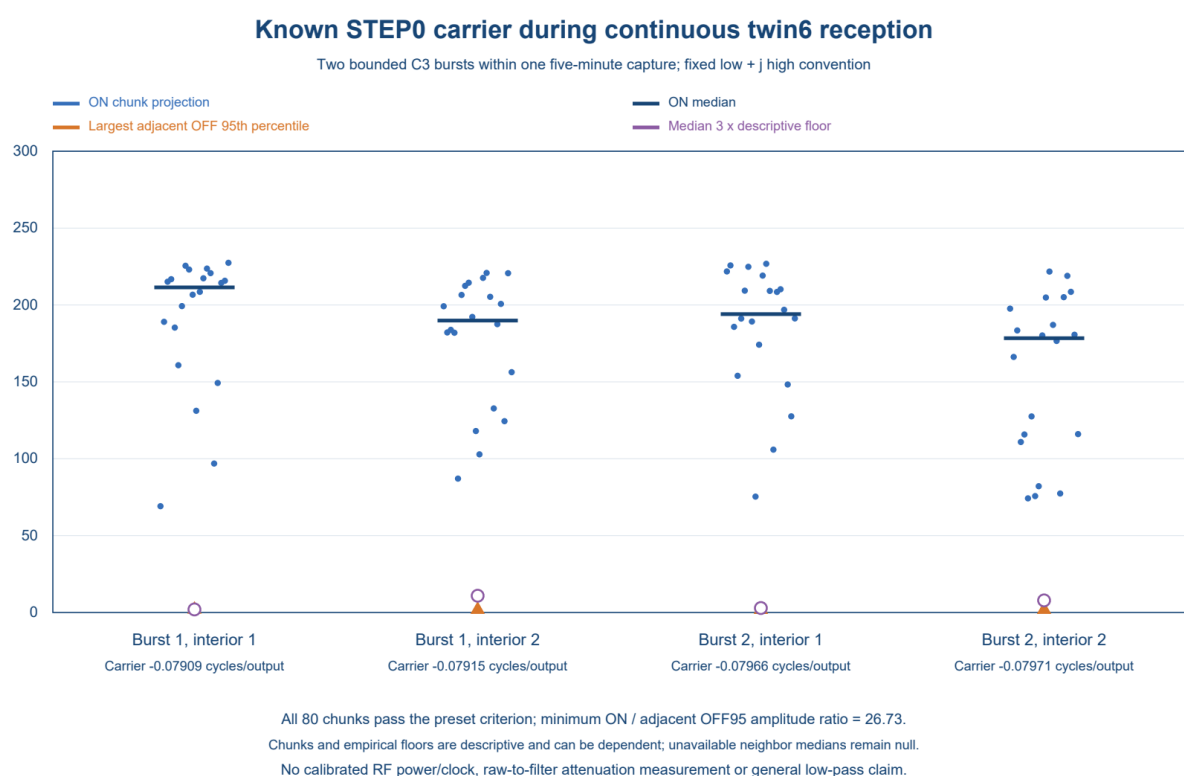


Figure 5. Reviewed aggregate source-linked amplitude/frequency plot from the two bounded STEP0 bursts. OFF controls and preset acceptance remain distinct from model-dependent descriptive floors. This contains no original waveform samples.

Saved Pi host monotonic timestamps put the first userspace frame 6.931463 ms after reader arm and 6.515833 ms after the private query starts. Maximum userspace interframe gap is 1.458134 ms; STOP to genuine vendor completion is 2.122691 ms. These are host acquisition/transport latencies, not RF propagation delay, ADC timing or a calibrated radio clock. The latency supplement does not repeat the full saved-frame audit. Known-signal aggregates

8c Longer passive twin6 recordings and counter rollover

The accepted ten-minute ABI36 follow-up delivers 488,320 checked frames containing 500,039,680 complex words over 600.051726406 host seconds, using b4b firmware/d437 driver. The separate accepted ABI37 one-hour follow-up delivers 2,929,920 checked frames containing 3,000,238,080 complex words over 3600.316538804 host seconds, using the coherent-read 58a firmware and matching 44c driver. Both retain thirty-two publication slots, the same twin6/native1.5 filter leaf and the timing, ownership, lag, reader-renewed lease, watchdog and normal STOP guards. The ABI37 change concerns bounded clock-read coherence and first-failure evidence, not filter or transfer performance.

ABI37 uses a bounded coherent clock read and retains diagnostic state for a rejected reading. Saved observations witness one clock high-word advance during the hour. They do not establish whether the bounded retry branch executed.

Full saved-file/frame, wire, normal STOP, paired-control restoration, newest 128 stopped-tail, archive and retained-health audits pass for each run. The hour records 22890 native watchdog services and 22890 reader renewals; its initial-plus-renewed heartbeat is 22891. Its complete payload sequence and publisher checksums pass. The two trials are passive, with no new known-source burst or protocol decode on their reduced-rate output. The historical source demonstration in section 8b remains a separate five-minute RF result.

The raw-position metadata crosses the uint32 boundary once in the ten-minute recording and 11 times in the hour recording. The audit reconstructs bounded positive modular differences and checks the terminal position against selected output count and the full reported ring-wrap/pointer position. This establishes the observed rollover under the native bounded-progress contract; it does not independently exclude unobserved whole source-ring laps or calibrate sample time. The hour's 3,000,238,080 selected words exceed signed32 range while remaining below unsigned32 rollover. Selected-word-counter rollover is therefore a separate, untested boundary.

One accepted hour is a longer tested operating point, not unlimited reliability or a systematic CPU/storage stress result. No original ambient waveform, operational controller, private network detail or decoded payload is added to the public package. Long-duration and finite-reception aggregates

9 Related work and reproducibility

Pinned primary-source comparison distinguishes the contribution from existing work. Nexmon SDR commit 70733afc documents arbitrary time-domain transmission, while Nexmon CSI commit 1232edf2 explicitly supports Pi 5 per-frame frequency-domain CSI. Pair-Fi commit 26d57310 documents finite BCM4389c1 raw reception followed by on-chip decoding. ESP-SDR commit fb264f96 documents continuous filtered S3 streaming and S31 DMA/ring ownership. Its continuous S3 PR 4 merged 2 October 2026 at 17:10:43 UTC. Continuous Wi-Fi-chip reception, generic reusable rings and integrity checks are therefore established ideas. We make no broad first-device, first-Pi5, priority or head-to-head performance claim. The bounded comparison concerns this exact BCM43455c0 selected-output implementation, private control lifecycle and measured operating points. Commit timestamps alone do not prove first public disclosure. Pinned comparison and line witnesses, ESP-SDR PR 4, Nexmon CSI Pi5 discussion

The historical source recipe at src/device reproduces the tested ABI36 b4b image and complete d437 module. The separate ABI37 recipe at src/device-abi37 reproduces the complete 652,800-byte 58a image and 656,600-byte 44c driver module with the documented pinned inputs and toolchains. The fresh unprivileged standalone build matches the full module SHA and all 45 allocated/relocation sections; it installs nothing. Native ELF debug and symbol bytes can differ with paths even when the composed image is exact. These are software-build results; the acquisition and recovery controller remains private. Historical source guide, ABI37 source guide.

Manufacturer binaries and original ambient I/Q remain excluded. The tested full acquisition controller depends on private freshness, installation, recovery and health records and is withheld. The protocol, inspected mechanism, frame verifier and integration prerequisites are available; the hardware experiment is not turnkey. Public readers can reproduce the source builds, synthetic checks, aggregate calculations and available figures, but cannot repeat the complete historical wire/health/payload audit or run the full experiment from this package alone. The owner selected

GPL-2.0-or-later for original code and CC BY 4.0 for the original report and diagrams. Existing upstream terms remain applicable. See licence grants and exclusions.

The findings are supported by saved experiment evidence and source-bound analysis, with extensive AI assistance. Human contributions and AI assistance are described in the accompanying contribution statement. No external hardware replication or peer-review status is claimed.

10 Conclusion and continuing disclosure

The demonstrated achievement is experimental radio-sample streaming from the Pi 5's existing Wi-Fi chip, with a checked one-hour recording. It shares ESP-SDR's broad aim of making receive waveforms available to software. The specific contribution here is sustained, checked BCM43455c0 export from an existing diagnostic capture mechanism; a practical general-purpose receiver remains a further research stage.

The accepted implementation delivers sustained selected complex receive output from one active BCM43455c0 collector, with checked reusable-memory publication and concurrent SDIO host retrieval. The original and later five-minute results establish separately bounded operating points for selected and partially filtered export. The later ten-minute and one-hour passive audits extend duration and observed counter-rollover validation, with signal characterisation evaluated separately. Finite raw epochs show modest additional selected-line attenuation for twin6. The STEP0 recording demonstrates two known-source bursts during a separately checked five-minute stream. Exact firmware and module rebuilding strengthens implementation reproducibility without providing a turnkey hardware controller. Complete-input acquisition continuity, calibrated RF performance and comprehensive anti-alias filtering remain open.

This report is intended as useful technical disclosure and a foundation for further research. Its purpose includes an accessible dated record for prior-art consideration, without a guarantee about future patents or third-party rights. Its eventual public date will be recorded when enabling content actually becomes available. UK novelty provisions, EPO enabling-disclosure guidance

The working overview can evolve throughout the research. Each released snapshot retains its files, evidence cutoff and actual public-availability record. New teaching, corrections and additional experiments receive their own dated version. A DOI identifies that record and does not certify technical correctness or novelty. Zenodo versioning

References and evidence

- Public run registry and exact tested identities, derived from the retained complete audits named there.
- Claim evidence table, linking report propositions to public artifacts and stated limits.
- Available receipt bins, derived from real userspace timestamps.
- Implementation appendix, extracted from inspected firmware/driver semantics.
- Matthias Schulz, Daniel Wegemer and Matthias Hollick. Nexmon: The C-based Firmware Patching Framework, 2017. Project.
- Francesco Gringoli, Matthias Schulz, Jakob Link and Matthias Hollick. Free Your CSI: A Channel State Information Extraction Platform For Modern Wi-Fi Chipsets, WiNTECH 2019. DOI.
- SEEMOO. Nexmon software-defined radio project, MobiSys 2018 repository. Source.
- ESPARGOS and contributors. ESP-SDR repository and continuous-I/Q documentation, retrieved 5 October 2026. Source.
- SD Association. Bus speed overview, retrieved 5 October 2026. Source.

Exact historical experiment/source hashes are retained in the private provenance inventory and public run registry. The pinned primary-source comparison records bounded inspected precedents; it is not an exhaustive literature search or a novelty certification.

Implementation appendix

This appendix describes the inspected V26 selected-stream implementation and its six-input append. The short sparse follow-up shares the publication ABI but has different input offsets and a channel-specific timing pair. The descriptions below are source-equivalent explanations; the included source-only recipe exactly rebuilds the later twin6 image/module; manufacturer input terms and the withheld operational controller still limit complete public hardware reproduction.

A Compatibility and ownership

The implementation is bound to BCM43455c0 manufacturing firmware 7.120.5.1 C0 WLTEST, the patched V26 image layout and a specific Linux brcmfmac module. The module checks the chip/revision, SDIO state/clock, code fingerprint, owned-RAM descriptor and single-session quotas. A matching chip family name alone does not admit another image.

The preserved upstream MFG base image is 507,048 bytes, SHA-256 a8793f02aed5730fc8020e93e49911e6183458497e0b26be633a2e35d337ab8a, identified in the project at Nexmon revision 68720406f54ff16cad28fd3c51f2392aefad9fb7. The identified path is firmwares/bcm43455/7_120_5_1_sta_C0/bcmdhd_mfg.bin_blob. This is an acquisition lead, not a verified redistribution permission. Identified upstream location

The experimental sample workspace is outside vendor heap ownership after the reviewed heap/CRC adjustment. The original first-32-byte text fingerprint and fixed record addresses are preserved by the six-input append. Its assembly accesses the record at 0x216980. These addresses are image-specific and are not general chip register specifications. Cache admission requires observed SCTL data-cache bit 2 clear; the implementation does not silently reconfigure caching to satisfy that condition.

The owned record occupies 33,664 bytes: a 128-byte descriptor, eight 4,128-byte slots and a 512-byte stopped reference. Other progress, context, phase, lease and watchdog workspace is additional. A complete build must retain native ROM helper resolution, hook redirects, linker ownership, image CRC/heap changes and the matching driver contract.

A1 Baseline identities and collector addresses

The source collector extent is 0x8000 through 0xBFFF in word-address units, comprising 16,384 words or 65,536 bytes. The firmware converts a word position into the selector byte address by shifting left two bits. It accesses selector/data ports at offsets 0x130 and 0x134 from the verified context. The sample collector's physical tap and undocumented clock circuitry remain qualified.

The unfiltered V26 image SHA-256 is 69cef337c50de6a2cce6a7402c919c74b3fed973b14bf19996a8f52fc515b4e. The six-input image SHA-256 is 07ab31ff7cd78a682f493ece9e3923d58a1d1a9dbf288d31c48cc89eface2dac. Their shared private-stream module SHA-256 is be069e1217374b74679f1a53515c7d7f634c324a46a54c8606b85cb12ece1301. The registry also identifies the different lease and sparse configurations. Firmware and host-driver compatibility are part of the mechanism, not interchangeable environmental details.

B Collector request and lifecycle

The fixed request uses the 48-byte version 2 sample_collect ABI with trigger 1, timeout 2, mode 17, nominal pre/post 400/400 and otherwise zero reserved fields. These finite request fields do not describe the long capture duration. The patched collector polling/stop/lease lifecycle determines the sustained experiment's end.

Stage	Behaviour and evidence requirement
Admission	Exact arguments, context, fingerprint, idle session, initial heartbeat and nonzero native watchdog predicates pass

Stage	Behaviour and evidence requirement
Preparation	Initialise owned descriptor and state; save native controls/selectors; apply and verify the chosen full timing pair where relevant
Active collection	Observe writer progress; consume completed inputs; commit output blocks; check stop/lease and service native watchdog
Stop or expiry	End collection through the reviewed native path at block alignment; separately account for any unread suffix
Stopped reference	Read latest128 with the same selected/filter kernel; compare against the latest retained firmware slot
Restoration	Preserve native result, restore/read back full changed pair and selectors, export final telemetry
Refusal or fault	Preserve evidence; use the separately bounded recovery contract rather than assuming successful normal restoration

The six-input wrapper performs native deadman/heartbeat admission before pair mutation. It admits the channel 40 ordinary full pair 0x19ce38e3, writes the requested 0x1ab55554 through the native low-then-high helpers, checks readback and pre-arm state, calls the original target, checks the pair after its stopped reference, then restores the baseline. Error results are preserved. The sparse channel 1 pair is different and must not be inferred from channel 40 literals.

C Source ring accounting

The hardware word extent is 0x8000..0xBFFF. Its relative writer progression is modulo 2^{14} . Successive snapshots derive $dw = (\text{pointer} - \text{current_previous}) \& 0x3fff$ and a modular cycle-counter delta. The inspected source rejects zero or excessive cycle delta above 32768 and observed word advance above 4096. It also requires compatible stopped/active extent, pointer-high, control, clock and context state.

produced extends the observed writer progression modulo 2^{32} . The next input coordinate starts at 64 and follows $\text{next_raw} = 64 + 16 * \text{output_count}$, with modular arithmetic where the reviewed helpers require it. The completion margin is 64 positions. The guarded overwrite limit is 16320, which is 16384 minus 64.

Source-equivalent eligibility is:

```
d = (produced - next_raw) modulo 2^32
if d has bit31 set, or d <= 64: no completed work
if d >= 16320: refuse excessive lag
available = produced - 64 - next_raw
n = ceil(available / 16)
if n < 32: defer
n = min(n, 128, remaining_positions_in_output_block)
```

The consumer snapshots and checks clock/progress after its reads, rejects $\text{produced_first} \geq 16320$, checks exact selector restoration and bounds read cost. The active path also preserves the reviewed full-iteration bound of 32768. These conditions depend on their observation/timing assumptions. They do not independently eliminate an unseen complete hardware lap or pause.

The selected-read address is $(0x8000 + (\text{raw} \& 0x3fff)) \ll 2$, written to selector offset 0x130, followed by the data read at 0x134. Firmware saves the old selector and restores it through the native template helper. No sample-memory acknowledgement/backpressure is introduced.

D Signed sample processing

For each packed word, lanes are interpreted as signed 16-bit values. ARM SHADD16 forms the widened signed sum of each lane and arithmetic-shifts it right one bit. The corresponding integer operation is $\text{half}(a, b) = \text{floor}((a+b)/2)$, including negative odd sums. There is no saturation or narrowing overflow for an average of two signed 16-bit values.

The six-input tree is:

```
l = half(half(x0, x1), x2)
r = half(x3, half(x4, x5))
y = half(l, r)
```

Its ideal weights are $[1,1,2,2,1,1]/8$; intermediate rounding makes it different from one final rational weighted sum. The reference utility computes this tree separately per lane. The assembly performs the two lanes simultaneously and advances 16 source positions after each output.

The sparse tree uses inputs at offsets 0, 1, 8 and 9:

```
y = half(half(x0, x1), half(x8, x9))
```

Its ideal weights are $1/4$ at those four positions. Its exact signed result lies within one integer unit below the ideal mean. Both live and stopped-reference paths use the same arithmetic, so their agreement is a consistency check. Separate same-epoch full raw exports establish the arithmetic against independently reconstructed inputs.

The three paths retain a fixed starting input coordinate and stride. Descriptive Hz values follow nominal input/output-rate assumptions; they do not add clock calibration. The kernels are short prefilters rather than established full-band anti-alias low-pass filters.

E Descriptor and publication slots

The descriptor contains 32 little-endian 32-bit words. Its fields in order are:

```
magic, version, bytes, capture_sequence, status, flags, count, stride,
first_raw, next_raw, produced, last_pointer, last_cycle, wraps, batches, max_lag,
copy_ticks, max_copy_ticks, restored_count, tail_checked, tail_mismatched,
vendor_result, cleanup, published,
sctlr, steps, total_blocks, block_words, slots, stop_request, stop_observed,
watchdog_feeds
```

The V26 contract uses magic 0x5354524d, version 26, bytes 33664, stride 16, block_words 1024, slots 8 and total_blocks 0xffffffff as a sentinel bound. Active status is 0 and clean stopped status is 2. Fatal flags/status are refused. Header counters alone do not establish physical-input generation.

Slot byte offset	Field	Meaning
0	sequence u32	Zero while reusing; committed generation when valid
4	first u32	Output index of the first word
8	count u32	1024 for a complete publication
12	hash u32	Word-wise publisher checksum
16	produced_before u32	Observed extended raw progression before filling
20	produced_after u32	Observed progression after filling
24	copy_ticks u32	Summed read work for this block
28	flags u32	Zero for accepted data
32	1024 u32 words	Little-endian packed signed-lane payload

Slot selection is $(\text{publication_sequence} - 1) \% 8$. Firmware invalidates sequence, executes DMB SY, writes the payload and metadata, completes the hash, executes DMB SY, writes committed sequence, executes DMB SY and advances published. There is no host ACK holding a slot indefinitely. If the host is too slow, refusal protects interpretation rather than recovering discarded generations.

The publisher checksum starts at 2166136261 and folds each packed 32-bit word as $h = (h \text{ XOR word}) * 16777619) \bmod 2^{32}$. This differs from folding four separate bytes. The six-input live assembly seeds

at output-block offset 0 and retains the accumulator over partial batches. The original final block rescan is removed, but the checked final commit order is preserved.

F Host data frame and acceptance

The exported data frame is 4,176 bytes. Scalar fields and payload are little-endian for the tested host. The format is documented independently of C padding below.

Byte offset	Type	Field
0	u32	Magic 0x49514652
4	u32	Version 1
8	u32	Frame bytes 4176
12	u32	Kind 1 data frame
16	u32	Expected output sequence
20	u32	First output index
24	u32	Word count 1024
28	u32	Publisher hash
32	u64	Kernel acquisition start_ns
40	u64	Kernel acquisition end_ns
48	8 u32	Slot metadata copied from firmware
80	1024 u32	Payload

The kernel first checks the bound code fingerprint and 128-byte descriptor. It requires the expected version/layout, capture identity, stride, sentinel limits, status and flags. A published sequence behind the reader or more than eight blocks ahead is rejected. The kernel reads initial 32-byte slot metadata, a 4096-byte first payload, a 4096-byte second payload and final 32-byte metadata under SDIO host ownership, restoring its backplane window afterward.

Acceptance requires initial/final metadata equality, identical payload copies, expected sequence, exact first index/count, zero flags and checksum equality. Errors propagate to the file reader. The file is nonseekable and tracks partial-read offset within one frame; the acquisition worker assembles a complete frame before queueing it. The offline public verifier sees only the accepted serialized frame, so it cannot repeat the original double-copy predicate without those separate raw copies.

Kind 2 terminal frames and the 1232-byte stop summary have different semantics. The public data-frame parser explicitly refuses them rather than interpreting terminal metadata as sample payload.

G Watchdog and lease

The native watchdog retains the existing configured deadline of 481500000 and requires a nonzero service interval/handle, exact native handler identity and admitted core context. It invokes the existing refresh routine with argument 1, observes execution cost and verifies deadline/core restoration. No interrupt-mask or watchdog-disable operation is introduced.

The 64-byte lease record includes magic/version/size, host heartbeat, seen token, renewal count, expiry counterticks, last cycle, expired/status, age/max gap, flags and token-wrap telemetry. Initial heartbeat 1 is admitted. Each renewal is the exact next token with UINT32_MAX followed by 1; token 0 is not an accepted wrap. Expiry is half the existing watchdog deadline, 240750000 counterticks, whose conversion to wall-clock time depends on the actual counter.

The host renews after each 128 accepted blocks while capture is active. Firmware separately checks lease age using modular cycle arithmetic. A missing reader can trigger normal expiry and EOF in the preserved expiry

experiment. That result is distinct from the 60-second explicit-close run and is not reproduced by a synthetic token test.

H STOP and control exclusion

The close path checks live image/descriptor/work state, writes the fixed value 0x53544f50 to descriptor byte offset 116 and reads it back. It accepts no caller-selected address or value. The producer observes the owned word, reaches the reviewed stop boundary and returns through the vendor cleanup path.

The stop summary reads the latest published slot's last 128 words and the stopped reference and requires equality with zero descriptor/vendor/tail errors. One extra firmware block can be published after the requested host quota. Therefore the reference may match that unread suffix; comparing it to the last delivered host frame would be the wrong predicate.

The private control path requires CAP_SYS_RAWIO and one matching radio. A module-global admission counter enrolls all participating protocol-mutex callers. A private owner transitions gate 0 to its exclusive sentinel only when no earlier user is enrolled. Other controls return EBUSY; the owner issues one real sample_collect request through the existing BCDC path. The reply is retained rather than fabricated early. This separates long-capture ownership from shared networking control.

I Host recording and guarded completion

An acquisition thread validates/assembles frames while a bounded writer queue performs buffered file writes. Queue errors, capacity exhaustion, exceptions or unexpected vendor completion cause refusal rather than a successful shortened recording. The original V26 five-minute runs use 8,192 frames; short tests can have smaller bounded queues and must retain their actual identities.

Buffered writes can take longer than on-chip retention, so acquisition and storage are separated. Short write-call timing does not measure durable flush completion. The run's final queue drain and writer termination are checked, with an independent saved-file audit afterward. Error recovery retains the original evidence and verifies restored/retained health under its reviewed procedure. New public deployment wrappers require explicit validation status rather than inheriting historical success automatically.

J General method and porting constraints

The implemented combination is a circular diagnostic collector, bounded embedded sample consumer, selected/transformed output blocks, ordered reusable-memory publication, concurrent host-interface readout and checked bounded delivery. These abstractions map to the D11 sample ring, Cortex-R4 consumer, slot protocol, SDIO reader and userspace queue described above.

Another target would need demonstrated concurrent sample access, known memory ownership/cache semantics, sufficient service budget, verified host memory reads, control lifetime separation and recovery. Different ring sizes, block lengths, strides, filters, output integrity modes or sinks require explicit arithmetic and bounds. A polyphase or CIC label does not remove required high-rate input reads. No other-board port, sample-memory DMA or arbitrary-bus implementation is established here.

K Public checks and remaining dependencies

The public verifier checks schema, exact sequence/index, timing ordering, metadata, flags and the word-wise checksum for supplied data frames. Synthetic fixtures test rejection paths. Public aggregate metrics and delivery bins regenerate the available result charts, while original waveform/wire/health evidence remains private.

The included source-only recipe records exact transitive build inputs, toolchains, source hashes and upstream notices and reproduces complete later twin6 firmware/module bytes. Manufacturer inputs must be acquired by the reader under applicable terms. Original project code uses GPL-2.0-or-later; retained upstream terms apply separately. The private operational controller remains withheld, so build reproduction is established while a turnkey hardware experiment is not. This scope limits any claim of complete public experimental reproducibility.

L Later thirty-two-slot partial-filter configurations

The original appendix sections A-K specify the initial eight-slot V26/six baselines. Later selected sparse32/twin6 trials have ABI35/ABI36 and thirty-two 4,128-byte publication slots. Their 132,736-byte record contains a 128-byte descriptor, 132,096 bytes of slots and a 512-byte stopped reference. Additional workspace and heap/stack admission are still required. Source/module identities are different; none of the earlier addresses constitute a public installation recipe for these versions.

The accepted later reader uses one payload copy surrounded by metadata checks and the publisher checksum. This changes the stability contract from the earlier two-identical-copies path. Guard, sequence, publication, stopped newest-owned-tail and health audits are retained, but copying once is not an independent freshness proof.

The twin6 exact signed kernel at offsets 0,1,2,8,9,10 is:

```
l = half(half(x0, x2), x1)
r = half(half(x8, x10), x9)
y = half(l, r)
```

Its ideal weights are [1,2,1,1,2,1]/8 at those offsets. Intermediate negative rounding remains part of its actual semantics. Both live and stopped paths use the same kernel; independent finite raw-reference comparisons validate the signed tree separately. The public offline arithmetic utility currently implements the earlier six/sparse kernels; this section specifies the later kernel without implying that the synthetic fixture verifies its device implementation.

M Exact source-build reproduction and integration boundary

The source-only package includes the transitive native append and minimal kernel reconstruction. A reader supplies the manufacturer image and pinned kernel inputs under their applicable terms. Fresh builds reproduce complete b4b and d437 bytes; the driver comparator also checks 45 allocated/relocation sections, vermagic, srcversion, licence declaration and native fingerprint. No install occurred. Toolchain/header details and receipts are included in the source guide.

The operational controller remains private. A public runner still needs fresh-session admission, exact module/firmware loading, Ethernet management, captured state/rollback, quotas, lease renewal, exclusive control, recording, genuine completion and health checks. The source package is sufficient to reproduce builds; it is not sufficient alone to reproduce the complete hardware experiment. No new untested runner is implied.

N Duration extension, clock coherence and identities

ABI37's src/device-abi37 recipe reproduces the exact 652,800-byte 58a image and 656,600-byte 44c module; src/device preserves ABI36. Unprivileged builds match the full module SHA and all 45 allocated/relocation sections without installing. Native ELF debug/symbol bytes may vary with paths. Acquisition and recovery controllers remain private.

ABI37 uses bounded coherent clock acquisition across adjacent high-word carries and retains the first rejected-reading diagnostic. The accepted hour includes one observed clock high-word advance; execution of the bounded retry branch is not established by that observation.

The accepted ABI37 hour uses the same signed twin6 leaf and 32-slot layout as the ABI36 ten-minute predecessor. It delivers 2,929,920 frames/3,000,238,080 words over 3600.316538804 host seconds, with 11 observed raw-position uint32 crossings. Its selected-word count exceeds signed32 and remains below unsigned32 rollover; selected-word rollover is untested. Raw positions are wider reconstructions from bounded positive modular metadata and the exported source-ring wrap count/pointer. This does not supply an independent physical-generation counter or exclude hidden whole-ring laps. ABI37 adds one bounded high/low/high clock reread only on an adjacent carry and phase6 first-rejection evidence; other coherence and timing guards still refuse failures. Success rejects every phase6 observation. The matched driver contract changes fingerprints/ABI/addresses while retaining reader ownership, lease renewal, lag, timeout and STOP semantics. The full operational controller remains private.